

Introduction to R

Advanced Statistics Lab

Claas Heuer

Get R

From CRAN

RStudio

Ubuntu:

```
sudo apt install r-base
```

Some Introduction Material

From the developers: [Introduction to R](#)

RStudio: [Documentation](#)

[R for Data Science](#)

Background

- R is a general programming language with a specialization for statistics.
- R is generally used through a REPL and interpreted live.
- Functional programming paradigm: all about objects and functions.
- Major appeal: packages, graphics, and scientific workflows.
- Major drawback: scripts can grow without structure.

Assigning And Manipulating Variables

Note: R has two assignment operators, `<-` and `=`.

```
x = 5  
print(x)
```

```
[1] 5
```

```
x
```

```
[1] 5
```

```
y = x  
print(y)
```

```
[1] 5
```

```
x = 3  
  
print(x)
```

```
[1] 3
```

```
print(y)
```

```
[1] 5
```

Working With Data Frames

```
dat = data.frame(  
  name = c("Adam", "Paul", "Elly"),  
  age = c(22, 43, 54)  
)
```

```
str(dat)
```

```
'data.frame':  3 obs. of  2 variables:  
 $ name: chr  "Adam" "Paul" "Elly"  
 $ age : num  22 43 54
```

```
names = dat$name  
ages = dat[, "age"]
```

```
str(names)
```

```
chr [1:3] "Adam" "Paul" "Elly"
```

```
str(ages)
```

```
num [1:3] 22 43 54
```

Subsetting Data Frames

```
dat[dat$name == "Paul", ]
```

```
  name age
2 Paul  43
```

```
dat[dat$age %in% c(22, 54), ]
```

```
  name age
1 Adam  22
3 Elly  54
```

```
dat$id = 1:nrow(dat)
dat[, "id_2"] = dat$id + 2
```

```
print(dat)
```

```
  name age id id_2
1 Adam  22  1    3
2 Paul  43  2    4
3 Elly  54  3    5
```

```
dat$id[dat$name == "Adam"] <- 12
print(dat)
```

```
  name age id id_2
1 Adam  22 12    3
```

Reading And Writing Data

```
dat = data.frame(  
  name = c("Adam", "Paul", "Elly"),  
  age = c(22, 43, 54)  
)  
  
write.table(dat, file = "my_data.csv", row.names = FALSE, sep = ",")  
  
dat_read = read.table(file = "my_data.csv", sep = ",", header = TRUE)  
str(dat_read)
```

```
'data.frame':   3 obs. of  2 variables:  
 $ name: chr  "Adam" "Paul" "Elly"  
 $ age : int  22 43 54
```


Vectors And Matrices

```
x = as.integer(1)
str(x)
```

```
int 1
```

```
is.vector(x)
```

```
[1] TRUE
```

```
y = rnorm(10)
str(y)
```

```
num [1:10] 0.161 0.397 -0.688 -0.259 1.106 ...
```

```
D = matrix(data = rnorm(5 * 5), nrow = 5, ncol = 5)
D
```

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	0.7818261	-0.9846848	-1.68313231	0.2720552	-0.2974026
[2,]	-0.2497865	-1.4219702	-0.54052719	1.1137353	0.8414808
[3,]	-0.5612766	-0.6969566	-1.07146811	0.1501556	-0.4995392
[4,]	1.5717090	0.7514258	1.25386655	0.3106957	0.6961961
[5,]	0.8953812	-1.4237623	-0.01767046	0.3594474	1.8671142

```
class(D)
```

```
[1] "matrix" "array"
```

Matrix Helpers

```
means = colMeans(D)
means
```

```
[1]  0.4875706 -0.7551896 -0.4117863  0.4412178  0.5215699
```

```
d = diag(D)
d
```

```
[1]  0.7818261 -1.4219702 -1.0714681  0.3106957  1.8671142
```

Data Exploration And Visualization

```
library(dlookr)
library(ggplot2)

data(iris)
str(iris)
```

```
'data.frame':  150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species      : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1 1 1 1 1 1 1 1
1 1 ...
```

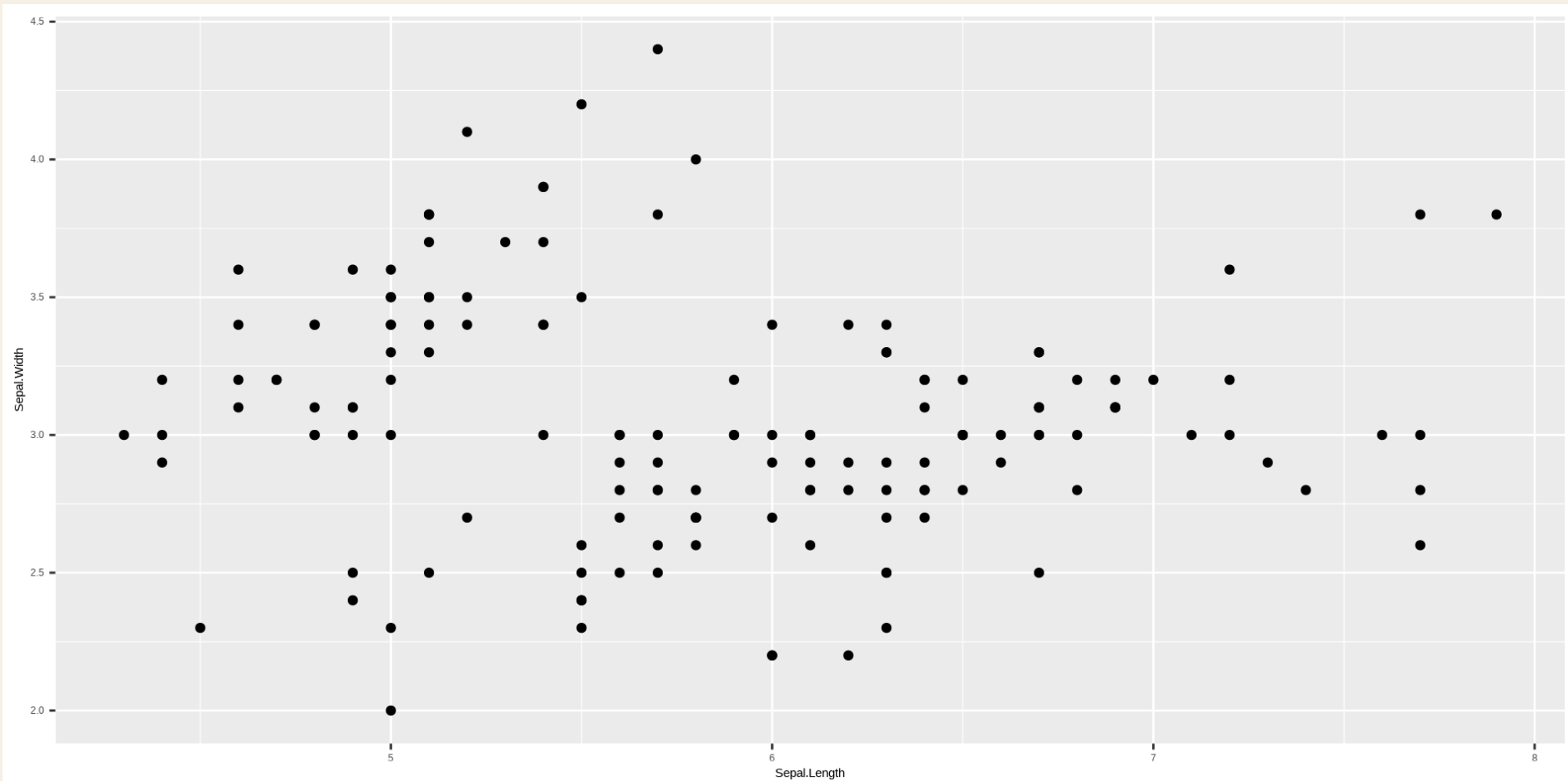
```
dlookr::diagnose(iris)
```

```
# A tibble: 5 × 6
  variables      types missing_count missing_percent unique_count unique_rate
  <chr>         <chr>         <int>          <dbl>         <int>         <dbl>
1 Sepal.Length numeric             0              0             35         0.233
2 Sepal.Width  numeric             0              0             23         0.153
3 Petal.Length numeric             0              0             43         0.287
4 Petal.Width  numeric             0              0             22         0.147
5 Species      factor             0              0              3         0.02
```

Scatterplot

```
p = ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width)) +  
  geom_point()
```

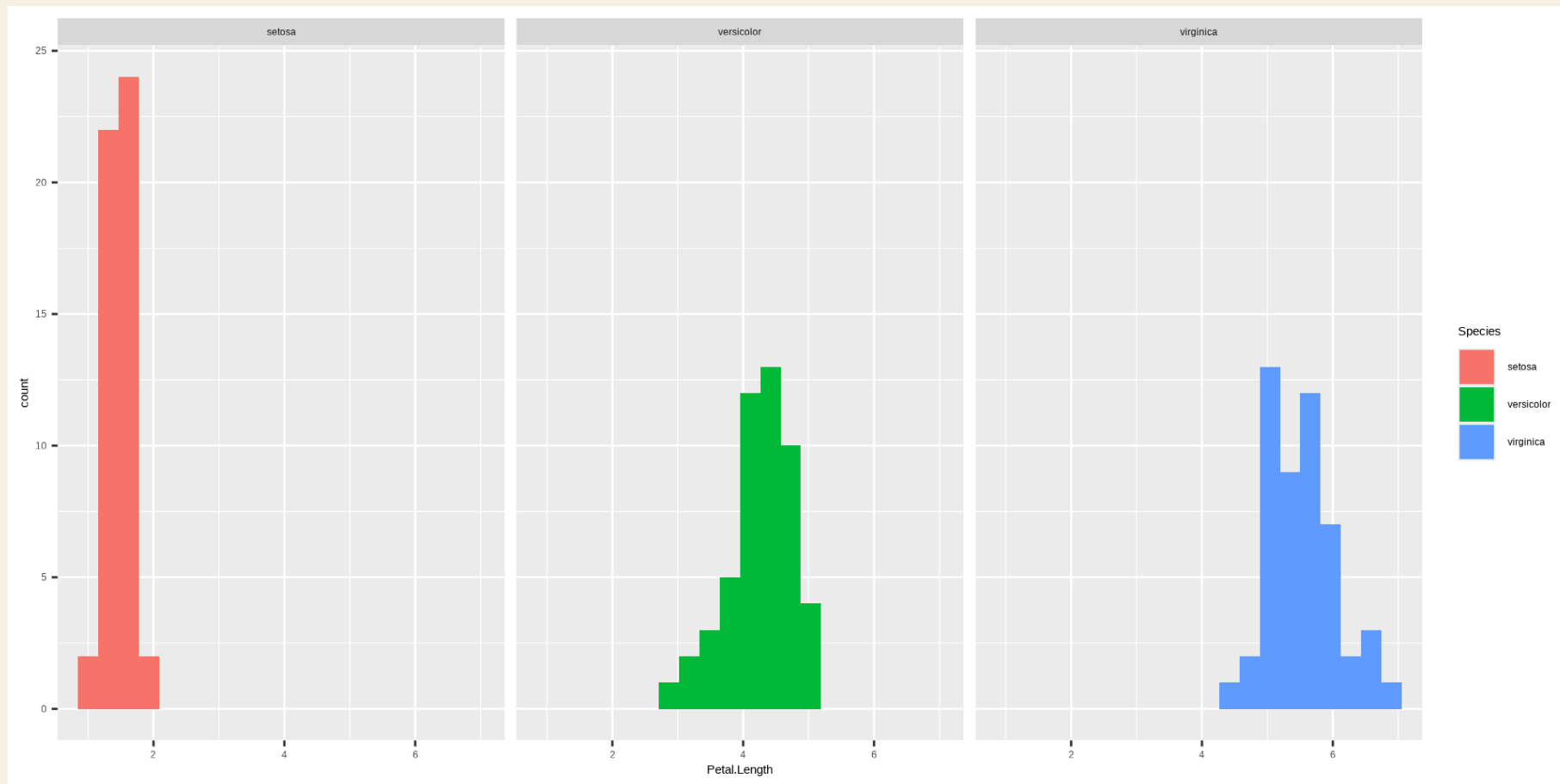
p



Histogram

```
p2 = ggplot(iris, aes(x = Petal.Length)) +  
  geom_histogram(aes(fill = Species), bins = 20) +  
  facet_wrap(~ Species)
```

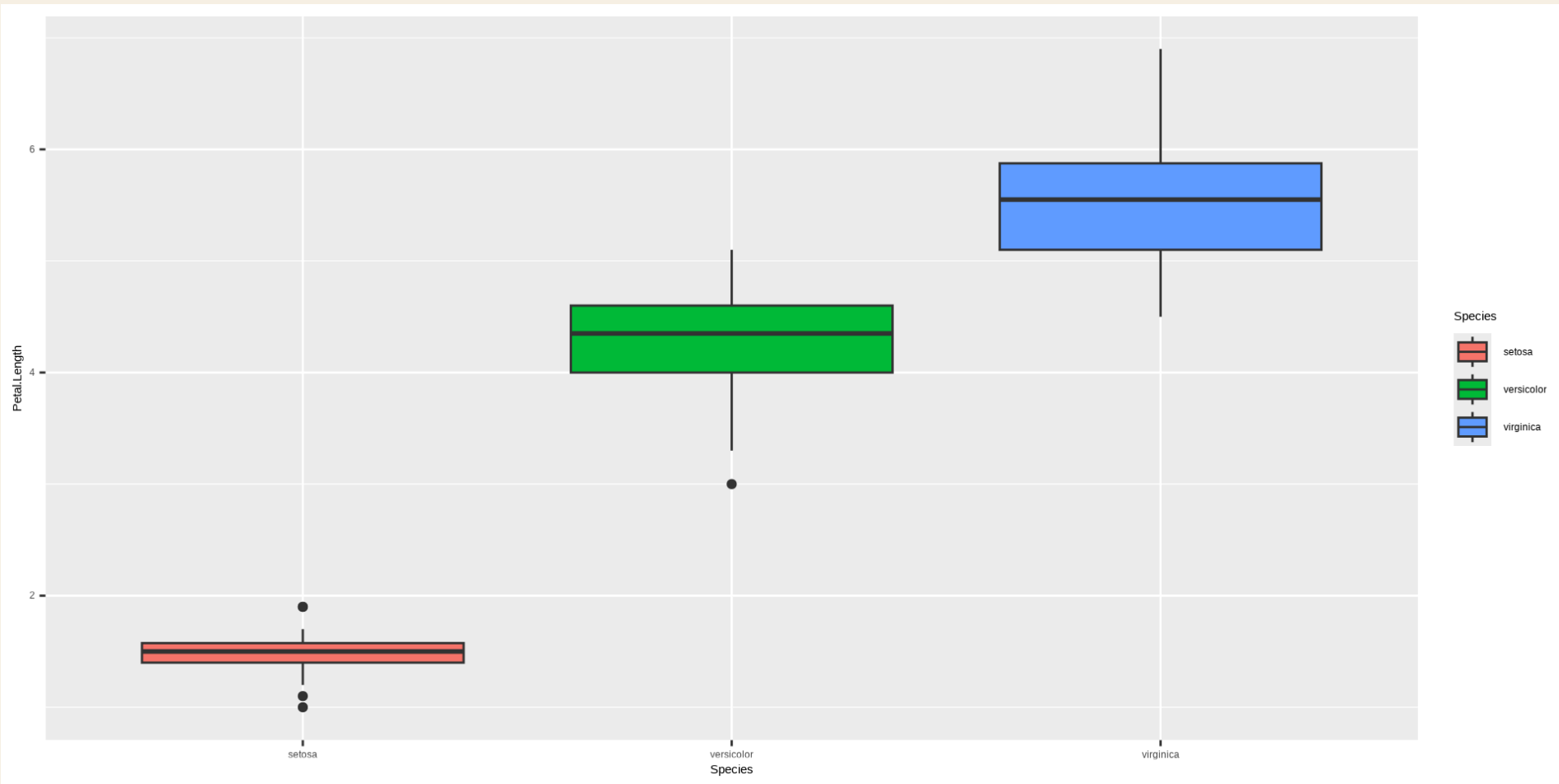
p2



Boxplot

```
p3 = ggplot(iris, aes(x = Species, y = Petal.Length, fill = Species)) +  
  geom_boxplot()
```

p3



Writing Functions

All operators in R are functions.

```
?sum  
?`+`  
fix(`+`)
```

A Mean Function

```
my_mean_function = function(y) {  
  n = length(y)  
  sum = 0  
  for (i in 1:n) sum = sum + y[i]  
  
  mean = sum / n  
  
  return(mean)  
}
```


Use The Function

```
x = rnorm(1000, mean = 23.42, sd = 4)
```

```
our_mean = my_mean_function(x)  
our_mean
```

```
[1] 23.59141
```

```
mean(x)
```

```
[1] 23.59141
```